

CHICAGO LEWIS, IL, USA

WMO: 725348

Lat: 41.604N

Lon: 88.085W

Elev: 201

StdP: 98.93

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 04831

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.1	-16.3	-23.8	0.5	-18.7	-21.0	0.6	-15.5	12.7	-0.7	11.7	-1.3	4.3	300	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	32.6	23.3	31.3	22.7	29.9	22.2	25.6	30.3	24.5	29.0	23.6	27.8	5.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.4	28.3	22.9	18.1	27.1	22.3	17.4	26.4	79.7	30.5	75.1	28.8	71.2	27.9	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.6	10.3	8.9	-23.5	34.8	3.5	1.5	-26.0	35.8	-28.1	36.7	-30.1	37.5	-32.7	38.6		
			-23.9	27.1	3.4	0.8	-26.3	27.7	-28.3	28.2	-30.2	28.6	-32.6	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	-4.4	-3.0	3.7	10.1	15.8	21.1	22.9	22.2	18.8	11.6	4.7	-1.5
	DBStd	10.94	6.57	6.38	6.22	5.33	5.05	3.57	2.99	2.88	4.20	5.18	5.31	5.94
	HDD10.0	1666	446	365	213	62	8	0	0	0	1	43	172	357
	HDD18.3	3423	704	597	454	251	113	13	2	3	45	218	408	614
	CDD10.0	1754	1	1	18	66	186	332	401	378	263	92	14	2
	CDD18.3	471	0	0	2	5	33	96	145	123	58	9	0	0
	CDH23.3	4171	0	0	11	71	331	866	1318	1007	487	79	0	0
	CDH26.7	1336	0	0	1	14	102	289	450	312	149	18	0	0
Wind	WSAvg	4.2	4.8	4.7	4.8	5.1	4.4	3.8	3.1	2.8	3.2	4.0	4.6	4.6
Precipitation	PrecAvg	910	46	41	56	91	106	107	98	100	78	74	65	48
	PrecMax	1153	103	143	128	176	187	251	196	243	320	201	149	125
	PrecMin	602	11	9	17	37	19	23	30	41	15	13	10	13
	PrecStd	126	25	27	28	38	41	55	39	41	60	48	39	27
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.2	16.2	25.0	28.3	32.2	33.8	34.9	33.9	32.7	29.0	21.1	16.2
		MCWB	12.4	12.0	17.6	16.9	20.6	22.4	24.4	25.5	22.8	21.6	14.5	13.7
	2%	DB	9.8	11.8	19.7	25.3	29.7	32.1	32.4	32.0	30.8	25.8	17.6	12.3
		MCWB	7.7	9.1	14.6	15.8	20.1	22.4	24.1	23.9	22.4	17.9	13.2	10.6
	5%	DB	6.7	7.7	16.4	22.5	27.3	30.0	31.0	30.0	28.0	22.7	15.5	9.0
		MCWB	5.0	5.4	12.3	14.9	18.8	21.6	23.6	22.9	20.4	16.8	12.2	7.5
	10%	DB	3.9	4.9	13.2	19.9	24.7	27.9	29.1	28.0	26.2	20.1	12.8	6.6
		MCWB	2.6	3.0	9.6	13.5	18.1	20.7	22.5	21.9	19.6	16.0	9.7	5.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.0	13.6	18.2	19.6	23.5	25.7	26.8	27.4	25.2	22.4	16.5	14.3
		MCDB	14.0	14.2	23.3	25.2	29.3	30.6	31.7	31.2	30.0	27.4	18.6	15.9
	2%	WB	8.0	9.5	15.7	17.5	21.6	24.1	25.5	25.6	23.5	20.1	14.4	10.7
		MCDB	9.0	11.7	18.9	22.9	26.7	28.9	30.6	29.8	27.7	23.0	16.8	11.6
	5%	WB	5.1	5.9	13.3	15.8	20.2	23.0	24.5	24.3	22.2	18.3	12.3	7.7
		MCDB	6.6	7.8	15.9	21.0	24.9	28.0	29.2	28.2	26.5	21.1	14.5	8.6
	10%	WB	2.7	3.1	9.9	14.1	19.1	22.0	23.6	23.1	21.1	16.3	10.2	5.0
		MCDB	4.0	4.6	13.2	19.1	23.2	26.6	27.9	26.6	24.7	19.6	12.8	6.5
Mean Daily Temperature Range	5% DB	MDBR	7.7	8.4	9.3	11.2	11.4	11.0	10.7	10.5	11.3	10.4	8.7	7.2
		MCDBR	9.9	11.3	13.3	15.7	14.0	12.8	12.2	12.3	13.2	14.0	12.1	9.8
		MCWBR	8.4	9.0	8.9	8.8	6.7	6.0	5.6	5.9	6.0	7.7	8.5	8.7
	5% WB	MCDBR	9.4	11.2	12.2	14.0	12.1	10.8	10.6	10.4	10.9	11.2	10.6	9.2
		MCWBR	8.5	9.4	8.9	9.0	7.0	6.1	6.0	5.9	5.7	7.5	8.6	8.7
Clear-Sky Solar Irradiance	taub	0.296	0.316	0.349	0.402	0.432	0.437	0.451	0.443	0.415	0.359	0.315	0.297	
	taud	2.423	2.372	2.374	2.236	2.208	2.235	2.194	2.209	2.258	2.414	2.489	2.449	
	Ebn at Noon	857	891	898	867	848	841	825	819	816	829	829	824	
	Edh at Noon	78	96	108	134	141	138	142	136	121	92	74	71	
All-Sky Solar Radiation	RadAvg	1.64	2.52	3.59	4.64	5.29	6.05	6.06	5.32	4.45	2.94	1.88	1.38	
	RadStd	0.17	0.33	0.27	0.38	0.44	0.42	0.32	0.31	0.38	0.32	0.18	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page